

Resampling Techniques

Problem 18: Bayesian Linear Regression

Goal:

- Implement Standard Gibbs vs. Blocked Gibbs Sampling
- Evaluate convergence using Trace Plots and ACF
- Analyze the impact of feature correlation on mixing

Model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon, \quad \epsilon \sim N(0, \sigma^2)$$

- Dataset provided with high correlation between x_1 and x_2

Prior Distributions:

- $\beta \sim N(\mu_0, \Lambda_0^{-1})$
- $\sigma^2 \sim \text{Inv-Gamma}(a_0, b_0)$

Defining the Blocks

In Gibbs sampling, we partition the set of all parameters Θ into disjoint subsets called **Blocks**.

The Partitioning

For our Linear Regression model, $\Theta = \{\beta_0, \beta_1, \beta_2, \sigma^2\}$.

- **Standard Gibbs:** 4 Blocks

$$B_1 = \{\beta_0\}, B_2 = \{\beta_1\}, B_3 = \{\beta_2\}, B_4 = \{\sigma^2\}$$

- **Blocked Gibbs:** 2 Blocks

$$B_1 = \{\beta_0, \beta_1, \beta_2\}, B_2 = \{\sigma^2\}$$

Key Idea: Larger blocks reduce autocorrelation but require sampling from Multivariate distributions.

How to Update Blocks (The Algorithm)

The update rule is: **Sample each block conditioned on the current values of all other blocks.**

For the 2-block version (Blocked Gibbs):

- 1 **Update Block 1 (β):** Keep σ^2 fixed at its last value. Draw a new β from:

$$P(\beta \mid \sigma_{old}^2, \mathbf{y})$$

- 2 **Update Block 2 (σ^2):** Keep the *new* β fixed. Draw a new σ^2 from:

$$P(\sigma^2 \mid \beta_{new}, \mathbf{y})$$

Iteration Rule

One full iteration consists of updating every block exactly once.

The Mathematical Updates

Step 1: Sampling the β Block

Using the current σ^2 , sample from the Multivariate Normal:

$$\beta \sim \mathcal{N}\left((X^T X)^{-1} X^T y, \sigma^2 (X^T X)^{-1}\right)$$

Step 2: Sampling the σ^2 Block

Using the current β , sample from the Inverse-Gamma:

$$a_n = n/2, \quad b_n = \frac{1}{2}(y - X\beta)^T (y - X\beta)$$

$$\sigma^2 \sim \text{Inv-Gamma}(a_n, b_n)$$

Implementation: The Conditionals

Standard Gibbs (Scalar Updates)

$$\beta_j | \beta_{\neq j}, \sigma^2 \sim \mathcal{N} \left(\frac{\sum x_{ij}(y_i - \sum_{k \neq j} x_{ik} \beta_k)}{\sum x_{ij}^2}, \frac{\sigma^2}{\sum x_{ij}^2} \right)$$

Blocked Gibbs (Vector Update)

$$\beta | \sigma^2 \sim \mathcal{N} \left((X^T X)^{-1} X^T y, \sigma^2 (X^T X)^{-1} \right)$$

Variance Update (Both)

$$\sigma^2 | \beta \sim \text{Inv-Gamma} \left(\frac{n}{2}, \frac{1}{2} (y - X\beta)^T (y - X\beta) \right)$$

Task 18.1: Implementation

Implement the following in Python:

- 1 **Run Standard Gibbs:** Update $\beta_0 \rightarrow \beta_1 \rightarrow \beta_2 \rightarrow \sigma^2$ (One by one).
- 2 **Run Blocked Gibbs:** Update $[\beta_0, \beta_1, \beta_2] \rightarrow \sigma^2$ (In groups).

Hyperparameters

$N_{\text{iter}} = 10,000$, Burn-in = 1,000, Thinning = 1

Task 18.2: Trace Plots and ACF

For both samplers, generate:

- 1 **Trace Plots:** Plot the value of β_1 against iteration number
- 2 **Autocorrelation Function (ACF):** Compute the correlation at various lags and plot them against lag k :

$$\rho(k) = \frac{\text{Cov}(x_t, x_{t+k})}{\text{Var}(x_t)}$$

Observation

Compare the "hairy caterpillar" appearance of the trace plots between the two methods.

Task 18.3: Convergence Diagnostics

- 1 Run 3 independent chains with different initializations
- 2 Compute the **Gelman-Rubin statistic** ($\hat{R}$):

$$\hat{R} = \sqrt{\frac{\widehat{\text{Var}}(\theta|y)}{W}}$$

Requirement

Confirm if $\hat{R} < 1.1$ for all parameters.

Problem 19: The Rosenbrock (Banana) Density

A "bad" density is one with high curvature or narrow ridges, where standard samplers get "stuck."

The Banana Function

$$f(x_1, x_2) \propto \exp \left(-(1 - x_1)^2 - 100(x_2 - x_1^2)^2 \right)$$

- The mode is at $(1, 1)$.
- The density follows a narrow, parabolic ridge.
- **Challenge:** High non-linear correlation. A fixed proposal width in MH will struggle to turn the "corners."

Implementation 1: Metropolis-Hastings (MH)

We use a fixed-width Symmetric Gaussian proposal:

$$q(\mathbf{x}^*|\mathbf{x}^{(t)}) = \mathcal{N}(\mathbf{x}^{(t)}, \sigma^2 \mathbf{I}).$$

The MH Loop Logic

For each iteration t :

- ➊ **Propose:** Generate $\mathbf{x}^* = \mathbf{x}^{(t)} + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$.
- ➋ **Acceptance Ratio:** Calculate $r = \frac{f(\mathbf{x}^*)}{f(\mathbf{x}^{(t)})}$.
- ➌ **Decision:**
 - Generate $u \sim \text{Uniform}(0, 1)$.
 - If $u < r$: Set $\mathbf{x}^{(t+1)} = \mathbf{x}^*$.
 - Else: Set $\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)}$ (**Keep the old value**).

Implementation 2: Component-wise Slice Sampling

Slice sampling updates one dimension at a time (x_1 , then x_2) by sampling from a horizontal "slice" of the density.

Step-by-Step for each Dimension ($d \in \{1, 2\}$)

- 1 **Vertical Slice:** Draw $y \sim \text{Uniform}(0, f(\mathbf{x}_{curr}))$.
- 2 **Stepping Out:** Start with interval $[L, R]$ of width w around x_d .
 - While $f(L, x_{-d}) > y$: $L = L - w$.
 - While $f(R, x_{-d}) > y$: $R = R + w$.
- 3 **Shrinkage:**
 - Sample $x_{prop} \sim \text{Uniform}(L, R)$.
 - If $f(x_{prop}, x_{-d}) > y$: Accept $x_d^{(t+1)} = x_{prop}$.
 - Else: Shrink the interval (L or R) to x_{prop} and re-sample.

Task 19.1: Simulation and Tuning

- 1 **Implement MH:** Run the sampler with three different proposal widths:
 - $\sigma^2 = 0.01$ (Tiny steps, high acceptance, slow exploration).
 - $\sigma^2 = 5.0$ (Large steps, low acceptance, "stuck" chain).
 - $\sigma^2 = 0.5$ (Attempt to find the "sweet spot").
- 2 **Implement Slice:** Run with initial width $w = 1.0$.
- 3 **Execution:** Run both for $N_{iter} = 10,000$.

Observation

Record the **Acceptance Rate** for MH. Note that Slice Sampling technically accepts a new point every iteration!

Task 19.2: Diagnostic Comparison

Compare the "Best" MH run against the Slice Sampler:

- 1 **Scatter Plots:** Plot the samples over the density contours. Which sampler fills out the "arms" of the banana?
- 2 **Autocorrelation (ACF):** Plot the ACF for x_2 . Which method shows the correlation dropping to zero faster?
- 3 **Trace Plots:** Observe the mixing. Does the MH chain stay in one part of the banana for too long?

Analysis Question

Why is the "fixed" proposal width of MH a disadvantage when the density has regions of both high and low curvature?